

Time Series Assignment

20183119

April 2025

1 Introduction

This report is focused on mortgage interest rates in the Republic of Ireland between 2014-2024. The data can be found here <https://data.gov.ie/dataset/retail-interest-rates-mortgage-rates>. The dataset I am focusing on has 41 quarterly observations where the first 37 will be used for model building and predicting the next 4 observations. This will be compared with the actual 4 remaining observations. The column this project focuses on is "Buy-to-let properties, floating rate, standard or LTV variable, rates on outstanding amounts (%)". It is required to first check whether the series is a random walk or white noise. This is done by various ACF,PACF and ADF checks.

2 White Noise/Random Walks

To check whether a series is white noise the series must have a zero mean, constant variance and no autocorrelation at nonzero lags. In r, the acf function was used to check this and a plot was created to show whether the series was white noise or not.

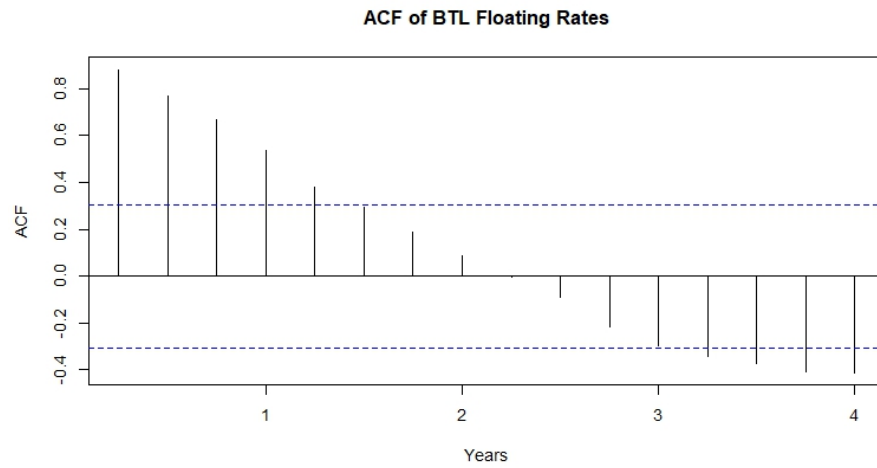


Figure 1: ACF Plot of full time series

Each vertical line shows a lag so we see that lags 1-5 are above the 95% confidence bands and lags 13-16 are below the confidence bands. If the series was white noise, then all time lags would lie within the 95% confidence bands. This shows that there is a clear structure to the data and the data we are working with is not white noise. Since the lags gradually get smaller it is fair to assume this series is not stationary but we will check more on this soon. A PACF test was also done

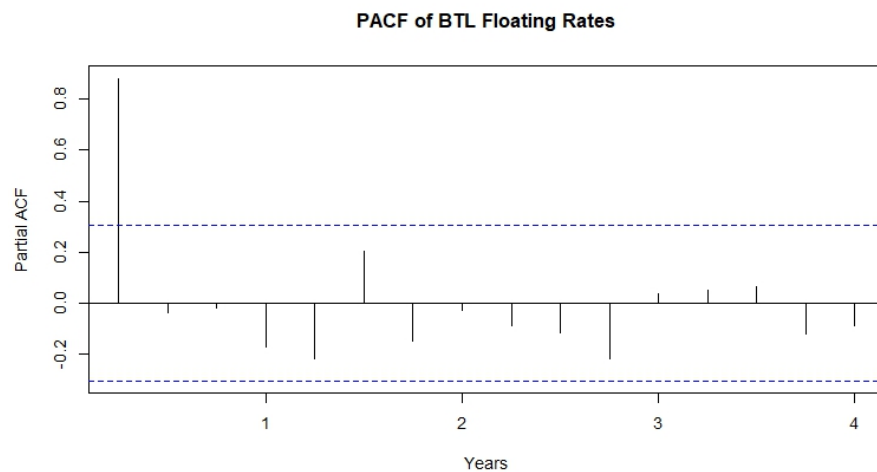


Figure 2: PACF of BTL Floating Rates

Figure 2 has one significant spike in the beginning followed by each lag within the 95% confidence bands. This shows how the series has an AR(1) feature. There is little linear correlation with older observations. This is a strong indication that the series is non-stationary.

To check if a time series is a random walk, we can check if it is non-stationary. To do this an augmented Dickey-Fuller test was performed. If the p-value is more than 0.05 we fail to reject the null hypothesis which is that the series is non-stationary. The results from this test were; Since the p-value was 0.98 we

Table 1: Augmented Dickey-Fuller Test for `ts_BTL_Floating_Rates`

Test statistic	-0.41739
Lag order	3
p-value	0.9807
Alternative hypothesis	Series is stationary

fail to reject the null hypothesis of a unit root which is consistent with a random walk.

Checking to see whether the series is a random walk on a seasonal level;

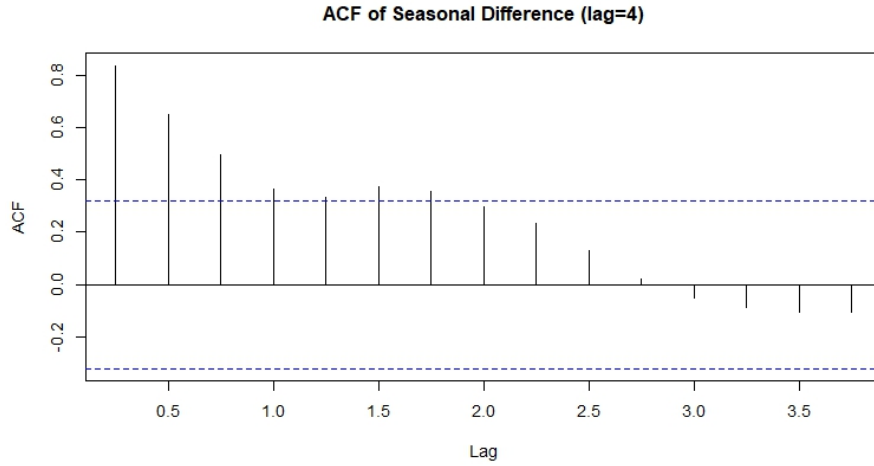


Figure 3: ACF of seasonally differenced series, lag 4

There are still many bands above the 95% confidence bands followed by a decrease in each lag. Hence, the series is not a seasonal random walk either.

3 Splitting the data 90/10

The data was then split where the first 90% of observations were used during the model building process that will later be tested on the final 10% of data.

This plot shows the first 90% of observations where the last 4 observations were removed.

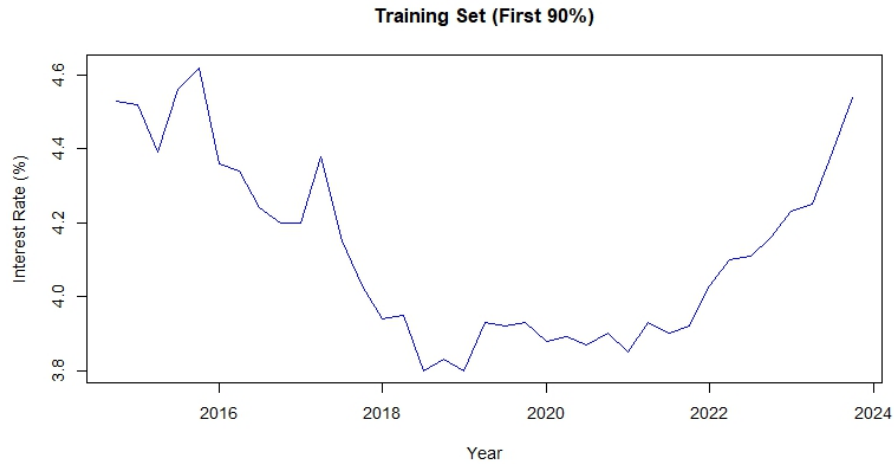


Figure 4: Plot of first 90% of data

These observations show a downward trend from 2016 until 2019, then the interest rates stabilise until just before 2022 where it shows a sharp upward trend until 2024. This is reflecting what happened with the European Central Bank's main interest rates hit 0% in March 2016. This was passed on to the mortgage rates which showed a decrease in figure 4. The market stabilised during the pandemic, the ECB held policy rates unchanged. Then the ECB began raising their policy in 2022 which is reflected by the sharp increase in interest rates seen in the figure. The data ends just before 2024 and will be used to test the models on.

4 Checking Stationarity

Due to the nature of the previous plot we can assume the dataset may not be stationary as the time series does not fluctuate around a constant mean but we can use an ADF (Augmented Dickey-Fuller) test to verify this. The output of this test is $Dickey - Fuller = 0.52427$, $Lagorder = 3$, $p - value = 0.99$

```
> adf.test(train_BTL)
```

Augmented Dickey-Fuller Test

```
data: train_BTL
```

```
Dickey-Fuller = 0.52427, Lag order = 3, p-value = 0.99
```

```
alternative hypothesis: stationary
```

A p-value of 0.99 is much higher than 0.05 means we fail to reject the null-hypothesis for non-stationarity and the Dickey-Fuller value of 0.52427 is close to zero which tells us the series is non-stationary.

To make the series stationary, it was differenced.

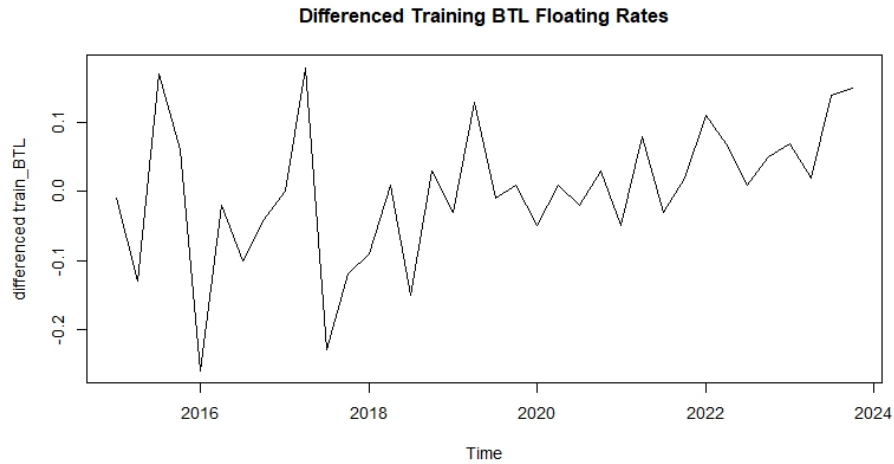


Figure 5: Plot of differenced first 90% of data

This plot shows the differenced series and it no longer has a clear trend. The line fluctuates around a constant mean which makes it look more stationary. This is ideal for building the ARIMA or SARIMA models. Running another ADF test to check stationary the results; *Dickey – Fuller* = -4.2266 , *Lagorder* = 3, *p – value* = 0.01231.

```
> adf.test(diff_train_BTL)
```

Augmented Dickey-Fuller Test

```
data: diff_train_BTL
Dickey-Fuller = -4.2266, Lag order = 3, p-value = 0.01231
alternative hypothesis: stationary
```

A Dickey-Fuller value of negative more than -3 is a good sign the series is stationary and combined with the p-value of less than 0.05 we can settle on the fact this differenced series is now stationary.

After differencing, another ACF and PACF test was performed;

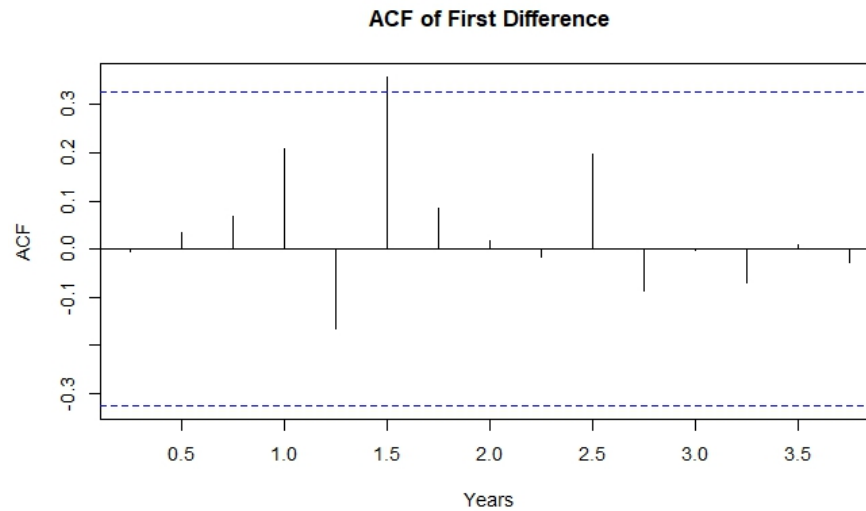


Figure 6: ACF of first difference

There is a spike showing strong positive autocorrelation at one lag. This shows an autoregressive component.

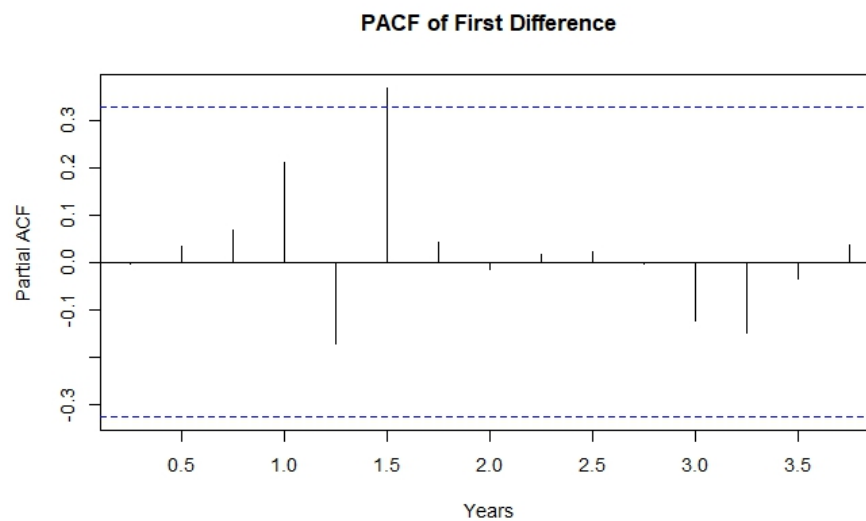


Figure 7: PACF of first difference

The PACF plot is similar with a single spike above the bands at the same

lag with the rest all within. This is a sign of an AR(1) process on the first difference.

For choosing the (p,d,q) values and since we have differenced once, d=1. The PACF spike suggests p=1 and the ACF trailing off after the spike indicates q=0. This points to a (1,1,0) ARIMA model for this data.

5 ARIMA Model

First, an ARIMA(1,1,0) model was built;

Table 2: ARIMA(1,1,0) Model Summary for Training Dataset (`train_BTL`)

Model	ARIMA(1,1,0)
Coefficient:	
AR(1)	−0.0046
Standard Error	(0.1697)
σ^2 (Variance)	0.01027
Log-likelihood	31.84
AIC	−59.67
AICc	−59.31
BIC	−56.51

Table 3: Training Set Accuracy Measures for ARIMA(1,1,0)

Accuracy Metric	Value
Mean Error (ME)	0.00038
Root Mean Squared Error (RMSE)	0.09857
Mean Absolute Error (MAE)	0.07287
Mean Percentage Error (MPE)	−0.0189%
Mean Absolute Percentage Error (MAPE)	1.7439%
Mean Absolute Scaled Error (MASE)	0.4580
ACF1 (Lag-1 Residual Autocorrelation)	0.00005

5.1 Residuals of ARIMA(1,1,0)

Table 4: Ljung–Box Test for Residuals of ARIMA(1,1,0)

Test statistic (Q*)	9.6883
Degrees of freedom	6
p-value	0.1384

As shown in table 4, the Ljung-Box p-value was 0.1384 confirms that no significant autocorrelation remains in the residuals.

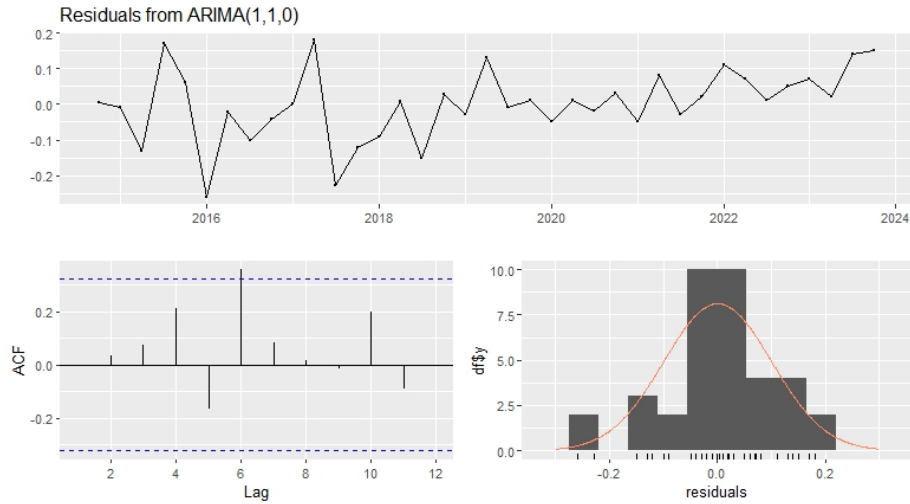


Figure 8: Residuals of ARIMA(1,1,0)

While in figure 8, the residual v time plot shows the residuals fluctuating around zero with a slightly decreasing variance over time. The ACF plot again shows a single spike above the 95% confidence bands indicating a single autocorrelation at one of the lags. The histogram shows an approximate normal distribution with no extreme outliers. All in all, this shows the ARIMA(1,1,0) residuals indicate a small amount of structure is left which contrasts the Ljung-Box results slightly. The Ljung-Box test overrules this case indicating the model is still appropriate for modelling.

Then this model was used to predict the next 4 observations.

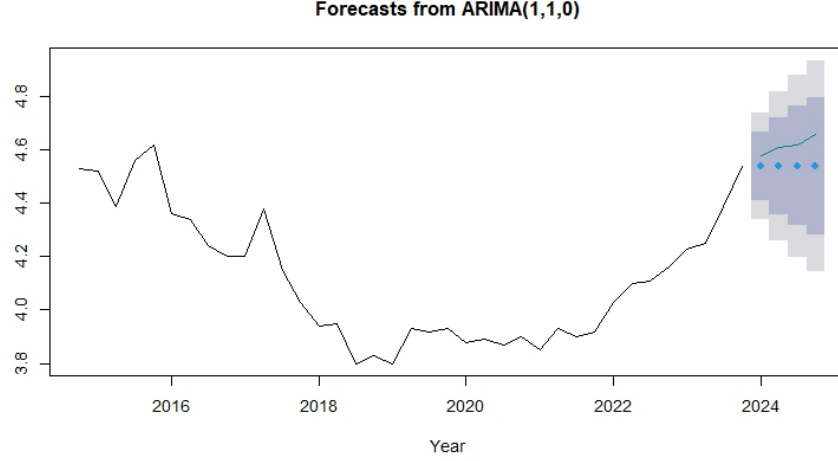


Figure 9: Forecast for ARIMA(1,1,0)

The ARIMA(1,1,0) model shows a flat path in 2024 somewhere in the middle of the confidence intervals. The actual observed observations show a more upward trend in comparison showing it does not capture the upward trend seen in the years leading up to it.

An ARIMA(0,2,1) model was fit to this series using the `autoARIMA` function in `r`, where 0 is the AR (auto-regressive) term, 2 is the integrated term and 1 is the MA term. This shows there was no auto-regressive term meaning that previous values do not necessarily influence future ones. The integrated term shows the model applied differencing twice to achieve stationarity indicating the original series had a clean trend that required differencing. There is 1 moving average in this model which means the current forecast errors are dependent on past forecast errors. The coefficient for this was -0.8605 showing a strong negative value means the errors fluctuate between positive and negative. The sigma squared value illustrated the variance of residuals in this model. The result of which was 0.01041 which shows errors are relatively small. An AIC score of -56.13 and BIC score of -53.02 are good negative values that indicate nice model fitting. The full results from the model are seen here;

The mean error is close to zero, the root mean squared error is small, the mean absolute value shows a good fit, the mean absolute percentage error shows excellent accuracy, the mean absolute scaled error is less than 1 which is also good. The lag-1 autocorrelation of residuals is -0.162 suggesting there is minimal leftover autocorrelation. These are very good error metrics to work with and we can tell the ARIMA model fits the historical data effectively.

Table 5: ARIMA Model Summary for Training Dataset (`train_BTL`)

Model	ARIMA(0,2,1)
Coefficients:	
MA(1)	−0.8605
Standard Error	(0.0804)
σ^2 (variance)	0.01041
Log-likelihood	30.07
AIC	−56.13
AICc	−55.76
BIC	−53.02

Table 6: Training Set Accuracy Measures

Accuracy Metric	Value
Mean Error (ME)	0.0160
Root Mean Squared Error (RMSE)	0.0978
Mean Absolute Error (MAE)	0.0703
Mean Percentage Error (MPE)	0.3767%
Mean Absolute Percentage Error (MAPE)	1.6883%
Mean Absolute Scaled Error (MASE)	0.4421
ACF1 (Lag-1 Residual Autocorrelation)	−0.1625

5.2 Residuals of ARIMA(0,2,1)

Table 7 showing the results of the Ljung-Box test for residuals for the ARIMA(0,2,1) model has a p-value of 0.067 meaning we fail to reject the null hypothesis of no autocorrelation in the residuals. This supports the idea that the model captured the series' structure.

Table 7: Ljung–Box Test for Residuals of ARIMA(0,2,1)

Test statistic (Q^*)	11.781
Degrees of freedom	6
p-value	0.0670

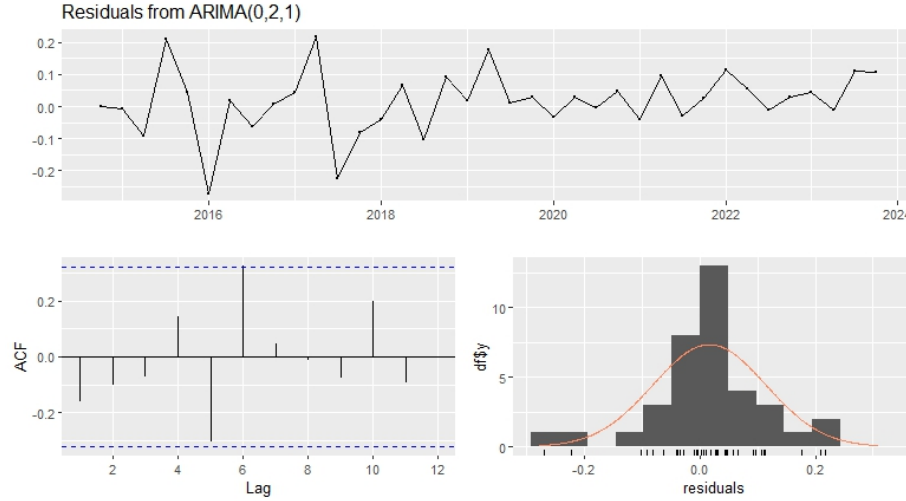


Figure 10: Residuals of ARIMA(0,2,1)

Figure 10's plot of residuals v time show a slightly more accurate fluctuation around zero with a slightly decreasing size of variance. In the ACF figure all lags lie within the 95% bands where lag 5 and 6 are close to being outside them. This indicates minimal correlation was leftover in the residuals. The histogram shows a normal distribution with no extreme outliers. Overall this shows the ARIMA(0,2,1) residuals are being treated as white noise and the structure of the series was captured.

Then we used the model to forecast the next 4 quarters so we could compare this with the actual data.

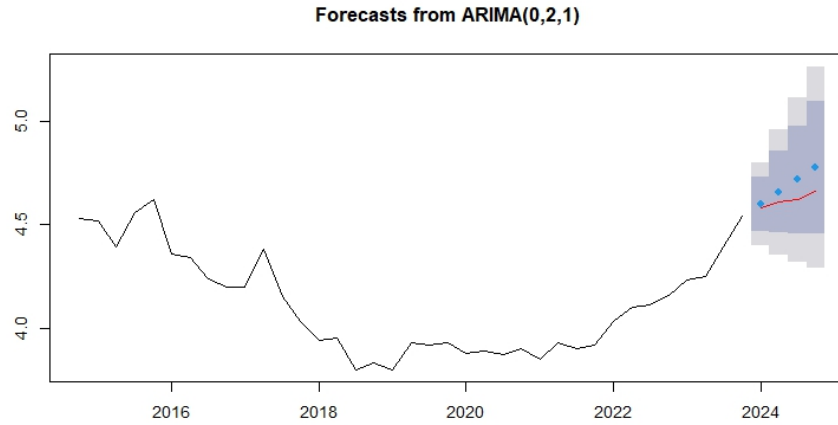


Figure 11: Forecasted values using ARIMA(0,2,1)

In this plot;

- The black line is our historical data.
- The blue dots indicate the forecasted values for the next four quarters.
- The shaded area is the confidence intervals showing uncertainty around the forecasted values.
- The red line indicates the actual observed data for 2024.

The forecasted blue dots capture the upward trend but slightly overestimate the increase compared to the actual data. It looks like the model of the prediction is practically accurate.

6 SARIMA Model

Since the dataset is quarterly data, a seasonal ARIMA model was fit to check whether there was a seasonal component to the data. The results of this model are seen below;

Table 8: SARIMA Model Summary (Quarterly Data)

Model	ARIMA(0,2,1)
Coefficients:	
MA(1)	-0.8605
Standard Error	(0.0804)
σ^2 (Variance)	0.01041
Log-likelihood	30.07
AIC	-56.13
AICc	-55.76
BIC	-53.02

Table 9: SARIMA Model Accuracy (Training Set)

Accuracy Metric	Value
Mean Error (ME)	0.0160
Root Mean Squared Error (RMSE)	0.0978
Mean Absolute Error (MAE)	0.0703
Mean Percentage Error (MPE)	0.3767%
Mean Absolute Percentage Error (MAPE)	1.6883%
Mean Absolute Scaled Error (MASE)	0.4421
ACF1 (Lag-1 Residual Autocorrelation)	-0.1625

This model actually chose the same ARIMA(0,2,1) as the non-seasonal one meaning there was no strong seasonal component found in the data. This model is identical to the previous ARIMA model so the results are the same.

7 Conclusions

The goal of this project was to model and forecast quarterly buy-to-let floating rates using Irish historical data from 2014-2024 using appropriate models. The ACF/PACF and ADF tests showed the series was non-stationary and not white noise. The first difference was stationary but still had some small structure to it confirming it was not a true random walk. While checking seasonality differencing and SARIMA auto-selection, no significant seasonal unit root was found or seasonal AR/MA terms. The best model choice for this dataset was the ARIMA(0,2,1) model that captured the upward trend of the observed data in 2024. This model had the lowest AIC score and sample errors (RMSE,MAPE) and the model passed all residual diagnostics.